

JEE Mains 2019 Chapter wise Question Bank

Limits – Questions

Q1

$$\lim_{y \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + y^4}} - \sqrt{2}}{y^4}$$

- (1) exists and equals $\frac{1}{4\sqrt{2}}$
- (2) exists and equals $\frac{1}{2\sqrt{2}(\sqrt{2}+1)}$
- (3) exists and equals $\frac{1}{2\sqrt{2}}$
- (4) does not exist

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Q2

For each $x \in \mathbf{R}$, let $[x]$ be greatest integer less than or equal to x . Then

$$\lim_{x \rightarrow 0} \frac{x([x] + |x|) \sin [x]}{x} \text{ is equal to:}$$

- (1) $-\sin 1$ (2) 1
- (3) $\sin 1$ (4) 0

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Q3

For each $t \in \mathbf{R}$, let $[t]$ be the greatest integer less than or equal to t . Then,

$$\lim_{x \rightarrow 1+} \frac{(1 - |x| + \sin |1 - x|) \sin \left(\frac{\pi}{2} [1 - x] \right)}{|1 - x| [1 - x]}$$

- (1) equals 1 (2) equals 0
- (3) equals -1 (4) does not exist

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Q4

Let f be a differentiable function such that $f'(x) = 7 - \frac{3}{4} \frac{f(x)}{x}$,

($x > 0$) and $f(1) \neq 4$. Then $\lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right)$:

- (1) exists and equals $\frac{4}{7}$. (2) exists and equals 4.
- (3) does not exist. (4) exists and equals 0.

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Q5

Let $[x]$ denote the greatest integer less than or equal to x . Then :

$$\lim_{x \rightarrow 0} \frac{\tan(\pi \sin^2 x) + (|x| - \sin(x[x]))^2}{x^2} :$$

- (1) does not exist (2) equals π
- (3) equals $\pi + 1$ (e) equals 0

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Q6

$$\lim_{x \rightarrow 0} \frac{x \cot(4x)}{\sin^2 x \cot^2(2x)} \text{ is equal to :}$$

- (1) 0 (2) 2 (3) 4 (4) 1

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Q7

Limits

$\lim_{x \rightarrow \pi/4} \frac{\cot^3 x - \tan x}{\cos(x + \pi/4)}$ is:

- (1) 4 (2) $4\sqrt{2}$
(3) $8\sqrt{2}$ (4) 8

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Q8

$\lim_{x \rightarrow 1^-} \frac{\sqrt{\pi} - \sqrt{2 \sin^{-1} x}}{\sqrt{1-x}}$ is equal to:

- (1) $\frac{1}{\sqrt{2\pi}}$ (2) $\sqrt{\frac{2}{\pi}}$
(3) $\sqrt{\frac{\pi}{2}}$ (4) $\sqrt{\pi}$

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Q9

$\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equals :

- (1) $4\sqrt{2}$ (2) $\sqrt{2}$ (3) $2\sqrt{2}$ (4) 4

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Q10

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying

$f'(3) + f'(2) = 0$. Then $\lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}}$ is equal

to :

- (1) 1 (2) e^{-1} (3) e (4) e^2

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Q11

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If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function and $f(2) = 6$, then

$\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{2t dt}{(x-2)}$ is:

- (1) $24f'(2)$ (2) $2f'(2)$
(3) 0 (4) $12f'(2)$

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Q12

If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is:

- (1) $\frac{8}{3}$ (2) $\frac{3}{8}$ (3) $\frac{3}{2}$ (4) $\frac{4}{3}$

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Q13

If $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$, then $a + b$ is equal to :

- (1) -4 (2) 5 (3) -7 (4) 1

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Q14

$\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$ is

- (1) 6 (2) 2 (3) 3 (4) 1

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Q15

Let $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$, $x \in \mathbb{R}$. If $f(x)$ attains maximum value at α and $g(x)$ attains minimum value at β ,

then $\lim_{x \rightarrow -\alpha\beta} \frac{(x-1)(x^2 - 5x + 6)}{x^2 - 6x + 8}$ is equal to :

- (1) $1/2$ (2) $-3/2$ (3) $-1/2$ (4) $3/2$

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Limits - Solutions

Q1

$$\begin{aligned}
 (1) \quad L &= \lim_{y \rightarrow 0} \frac{\sqrt{1+\sqrt{1+y^4}} - \sqrt{2}}{y^4} \\
 &= \lim_{y \rightarrow 0} \frac{(\sqrt{1+\sqrt{1+y^4}} - \sqrt{2})(\sqrt{1+\sqrt{1+y^4}} + \sqrt{2})}{y^4 (\sqrt{1+\sqrt{1+y^4}} + \sqrt{2})} \\
 &= \lim_{y \rightarrow 0} \frac{1 + \sqrt{1+y^4} - 2}{y^4 (\sqrt{1+\sqrt{1+y^4}} + \sqrt{2})} \\
 &= \lim_{y \rightarrow 0} \frac{(\sqrt{1+y^4} - 1)(\sqrt{1+y^4} + 1)}{y^4 (\sqrt{1+\sqrt{1+y^4}} + \sqrt{2})(\sqrt{1+y^4} + 1)} \\
 &= \lim_{y \rightarrow 0} \frac{1 + y^4 - 1}{y^4 (\sqrt{1+\sqrt{1+y^4}} + \sqrt{2})(\sqrt{1+y^4} + 1)} \\
 &= \frac{1}{2\sqrt{2} \times 2} = \frac{1}{4\sqrt{2}}
 \end{aligned}$$

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Q2

$$\begin{aligned}
 (1) \quad &\lim_{x \rightarrow 0^-} \frac{x([x] + |x|) \cdot \sin[x]}{|x|} \\
 &= \lim_{h \rightarrow 0} \frac{(0-h)([0-h] + |0-h|) \cdot \sin[0-h]}{|0-h|} \\
 &= \lim_{h \rightarrow 0} \frac{(-h)(-1+h) \sin(-1)}{h} \\
 &= \lim_{h \rightarrow 0} (1-h) \sin(-1) \\
 &= -\sin 1
 \end{aligned}$$

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Q3

$$\begin{aligned}
 (2) \quad &\lim_{x \rightarrow 1^+} \frac{(1-|x| + \sin(|1-x|)) \sin\left(\frac{\pi}{2}[1-x]\right)}{|1-x|[1-x]} \\
 &= \lim_{h \rightarrow 0} \frac{(1-|1+h| + \sin(|1-1-h|)) \sin\left(\frac{\pi}{2}[1-1-h]\right)}{|1-1-h|[1-1-h]} \\
 &= \lim_{h \rightarrow 0} \frac{(1-1-h + \sin h) \sin\left(\frac{\pi}{2}(-1)\right)}{h([0-h])} \\
 &= \lim_{h \rightarrow 0} \frac{(-h + \sin h) \sin\left(-\frac{\pi}{2}\right)}{h(-1)} = 0
 \end{aligned}$$

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Q4

Limits

(2) Let $y = f(x)$

$$\frac{dy}{dx} + \left(\frac{3}{4x}\right)y = 7$$

$$\text{I.F.} = e^{\int \frac{3}{4x} dx} = e^{\frac{3}{4} \ln x} = x^{\left(\frac{3}{4}\right)}$$

Solution of differential equation

$$y \cdot x^{\frac{3}{4}} = \int 7 \cdot x^{\frac{3}{4}} dx + C$$

$$y \cdot x^{\frac{3}{4}} = 7 \cdot \frac{x^{\frac{7}{4}}}{\left(\frac{7}{4}\right)} + C = 4x^{\frac{7}{4}} + C$$

$$y = 4x + Cx^{-\frac{3}{4}}$$

$$\Rightarrow f\left(\frac{1}{x}\right) = \frac{4}{x} + Cx^{\frac{3}{4}}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} x \cdot f\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} \left(4 + Cx^{\frac{7}{4}}\right) = 4$$

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Q5

(1) RHL is, $\lim_{x \rightarrow 0^+} \frac{\tan(\pi \sin^2 x) + (x-0)^2}{x^2}$

$$= \lim_{x \rightarrow 0^+} \left(\frac{\tan(\pi \sin^2 x)}{x^2} + 1 \right)$$

$$= 1 + \pi$$

And LHL is, $\lim_{x \rightarrow 0^-} \frac{\tan(\pi \sin^2 x) + (-x + \sin x)^2}{x^2}$

$$= \lim_{x \rightarrow 0^-} \frac{\tan(\pi \sin^2 x) + x^2 + \sin^2 x - 2x \sin x}{x^2}$$

$$= \pi + 1 + 1 - 2 = \pi$$

Since, $\text{LHL} \neq \text{RHL}$

Hence, limit does not exist.

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Q6

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(4) $\lim_{x \rightarrow 0} \frac{x \cot 4x}{\sin^2 x \cdot \cot^2 2x}$

$$= \lim_{x \rightarrow 0} \frac{x \cdot \tan^2 2x}{\sin^2 x \cdot \tan 4x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^2 \cdot \left(\frac{\tan 2x}{2x} \right)^2 \cdot \left(\frac{4x}{\tan 4x} \right) \cdot \frac{4}{2^2} = 1$$

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Q7

(4) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cot^3 x - \tan x}{\cos\left(x + \frac{\pi}{4}\right)} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\cot^3 x \left(1 - \frac{\tan x}{\cos^3 x}\right)}{\cos\left(x + \frac{\pi}{4}\right)}$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(1 - \tan^4 x)}{\tan^3 x \cos\left(x + \frac{\pi}{4}\right)}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(1 + \tan^2 x)(1 - \tan x)(1 + \tan x)}{\tan^3 x \left(\frac{\cos x - \sin x}{\sqrt{2}} \right)}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(1 + \tan^2 x)(1 + \tan x)(\cos x - \sin x)}{\frac{\sin^3 x}{\cos^2 x} \left(\frac{\cos x - \sin x}{\sqrt{2}} \right)}$$

$$= \frac{(2)(2)}{\frac{1}{(\sqrt{2})(\sqrt{2})}} = 8$$

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Q8

Limits

$$(2) \lim_{x \rightarrow 1^-} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1}x}}{\sqrt{1-x}}$$

$$= \lim_{h \rightarrow 0} f(1-h)$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1}(1-h)}}{\sqrt{1-(1-h)}}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{\pi} - \sqrt{2\sin^{-1}(1-h)}}{\sqrt{h}}$$

$$= \lim_{h \rightarrow 0} \frac{-\frac{1}{2\sqrt{2\sin^{-1}(1-h)}} \times 2 \times \frac{1}{\sqrt{1-(1-h)^2}} (-1)}{\frac{1}{2\sqrt{h}}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{2\sin^{-1}(1-h)}} \cdot \frac{1}{\sqrt{h(2-h)}}}{\frac{1}{2\sqrt{h}}}$$

$$= 2 \times \frac{1}{\sqrt{\pi}} \times \frac{1}{\sqrt{2}} = \sqrt{\frac{2}{\pi}}$$

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Q9

$$(1) \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{2\cos^2 \frac{x}{2}}} \quad \left[\frac{0}{0} \right]$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} \left[1 - \cos \frac{x}{2} \right]} = \lim_{x \rightarrow 0} \frac{\sin^2 x}{2\sqrt{2} \sin^2 \frac{x}{4}}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{\sin x}{x} \right)^2 \cdot 16}{2\sqrt{2} \left(\frac{\sin \frac{x}{4}}{\frac{x}{4}} \right)^2} = \frac{16}{2\sqrt{2}} = 4\sqrt{2},$$

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Q10

$$(1) I = \lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}} \quad [1^\infty \text{ form}]$$

$$\Rightarrow I = e^{\ell}, \text{ where}$$

$$I_1 = \lim_{x \rightarrow 0} \left(\left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} - 1 \right) \right) \left(\frac{1}{x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{1}{x} \right) \left(\frac{f(3+x) - f(3) - f(2-x) + f(2)}{1 + f(2-x) - f(2)} \right) \quad \left(\frac{0}{0} \text{ form} \right)$$

By L. Hospital Rule,

$$I_1 = \lim_{x \rightarrow 0} \left(\frac{f'(3+x) + f'(2-x)}{1} \right) \lim_{x \rightarrow 0} \left(\frac{1}{1 + f(2-x) - f(2)} \right)$$

$$= f'(3) + f'(2) = 0$$

$$\Rightarrow I = e^0 = e^0 = 1$$

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Q11

(4) Using L' Hospital rule and Leibnitz theorem, we get

$$\lim_{x \rightarrow 2} \frac{\int_2^{f(x)} 2t dt}{(x-2)} = \lim_{x \rightarrow 2} \frac{2f(x)f'(x) - 0}{1}$$

$$\text{Putting } x = 2, 2f(2)f'(2) = 12f'(2) \quad [\because f(2) = 6]$$

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Q12

Limits

(1) Given, $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow K} \left(\frac{x^3 - k^3}{x^2 - k^2} \right)$

Taking L.H.S. $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} \left(\frac{0}{0} \text{ form} \right)$

$\lim_{x \rightarrow 1} \frac{4x^3}{1} = 4$ [Using L Hospital's Rule]

$\therefore \lim_{x \rightarrow K} \frac{x^3 - k^3}{x^2 - k^2} = 4$

$\Rightarrow \lim_{x \rightarrow K} \frac{3x^2}{2x} = 4$ [Using L Hospital's Rule]

$\Rightarrow \frac{3}{2}k = 4 \Rightarrow k = \frac{8}{3}$

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Q13

(3) $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$

$\therefore$ limit is finite. $\therefore 1 - a + b = 0$

$\Rightarrow \lim_{x \rightarrow 1} \frac{2x - a}{1} = 5 \left(\frac{0}{0} \text{ form} \right)$ (By L Hospital's rule)

$\Rightarrow 2 - a = 5 \Rightarrow a = -3$ and $b = -4$

Then $a + b = -3 - 4 = -7$

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Q14

(2) Given limit is,

$\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$

On rationalising,

$= \lim_{x \rightarrow 0} \frac{(x + 2 \sin x) [\sqrt{x^2 + 2 \sin x + 1} + \sqrt{\sin^2 x - x + 1}]}{(x^2 - \sin^2 x) + (x + 2 \sin x)}$

$= \lim_{x \rightarrow 0} \frac{\left[1 + 2 \left(\frac{\sin x}{x} \right) \right] [\sqrt{x^2 + 2 \sin x + 1} + \sqrt{\sin^2 x - x + 1}]}{\left(x - \frac{\sin^2 x}{x} \right) + \left(1 + 2 \left(\frac{\sin x}{x} \right) \right)}$

$= \frac{3 \times 2}{3} = 2 \quad \left[\because \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right]$

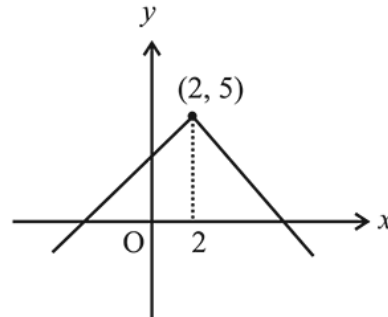
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Q15

(1) $f(x) = 5 - |x - 2|$

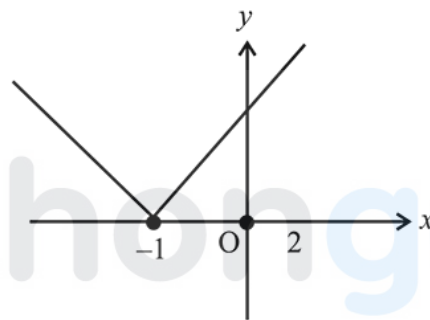
Graph of $y = f(x)$



By the graph $f(x)$ is maximum at $x = 2 \therefore \alpha = 2$

$g(x) = |x + 1|$

Graph of $y = g(x)$



By the graph $g(x)$ is minimum at $x = -1$

$\therefore \beta = -1$

Now, $\lim_{x \rightarrow 2} \frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)}$

$= \lim_{x \rightarrow 2} \frac{(x-1)(x-3)}{x-4} = \frac{1}{2}$

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